Sahaj Bhadja

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Summary

Early-career software engineer with 2+ years building **Python/C++-adjacent systems**, tested data workflows, and research-style tooling; strong in algorithms, distributed systems fundamentals, TDD habits, Linux, CI/CD, and quantitative problem solving.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Optimized **SQL-backed platform services** for 500+ users, tuning indexed access paths, background execution, and service logic to reduce dashboard latency from **1.2s to 350ms** under analytics workloads.
- Built **fault-tolerant validation and data workflows** processing 5,000+ monthly jobs, improving reliability through error handling, reconciliation checks, CI/CD releases, debugging notes, and operational documentation for analytics users.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Python and JavaScript services** handling 10,000+ daily requests, implementing concurrent file processing, REST APIs, persistence workflows, automated checks, and production bug fixes under sprint deadlines.
- Refactored **grading execution paths and test fixtures** around OCR, ingestion, and model outputs, documenting root causes while reducing instructor feedback turnaround by **60%** across repeated workflows safely.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Created **Python simulation and evaluation pipelines** across 100,000+ records, producing reproducible metrics, logs, and reports for model validation, benchmarking, debugging reviews, and experiment comparisons for researchers.
- Documented **algorithm behavior and edge cases** for researchers, turning ad hoc scripts into repeatable workflows with clearer handoffs, traceable outputs, reviewable experiments, and shared references during iteration.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform microservices and ETL workflows** using Python, SQL, AWS Glue, and S3, improving response time by **30%** while supporting distributed data movement and integrations safely and reliably.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito across validation flows, increasing CI reliability by **68%** and reducing regressions during coordinated releases and deployments safely.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **modular service features** with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting 2M+ synchronized records across distributed product workflows and data-heavy processes reliably under production constraints consistently.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving repeatable deployments, troubleshooting documentation, and team handoffs across shared environments and releases.

Skills

Languages: Python, C++, C, Java, SQL, Bash, JavaScript, TypeScript
Systems: Linux/UNIX, concurrency, scheduling, memory management, distributed systems, event-driven design
Testing: TDD, unit tests, integration tests, PyTest, Mockito, refactoring, debugging, benchmarking
Data: PostgreSQL, MySQL, MongoDB, ETL pipelines, time-series concepts, statistical evaluation, model validation
Delivery: Git/GitHub, CI/CD, Jenkins, Docker, code reviews, Agile/Scrum, technical documentation

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented **C++ scheduling algorithms** including FCFS, SJF, SRT, RR, priority, and MLQ with metric calculation and visualization across 1K+ process simulations, scenarios, and edge cases robustly and clearly.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a **containerized REST processing API** with AWS deployment, CI/CD releases, validation checks, and reproducible execution, achieving **87%** classification accuracy across evaluated inputs with reporting and diagnostics.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and database management through Java services and TypeScript interfaces for developers and analysts efficiently.